

IT PARK DEVELOPMENT ZONE ANALYSIS IN BANGALORE

GEOTA Project

Name : Saketh Pabbu

Roll No : S20230010169

INTRODUCTION :

Overview

The main aim of this project is to help IT companies find good spots for new offices in Bangalore without causing more traffic jams or overcrowding. We use pictures from space to check the city and create maps that show where IT parks already exist, where new ones are building, and where future ones could go. This way, planners can build better roads and metro lines, and investors can pick smart places to put their money. It's like a guide for growing Bangalore's tech scene in a smart way.

Zone Classifications

- **ESTABLISHED IT (Red):** Places with big IT parks already running, like Electronic City and Whitefield, close to roads and airports.
- **EMERGING IT (Yellow):** Areas building new IT stuff next to old parks, such as Sarjapur and Devanahalli.
- **POTENTIAL IT (Orange):** Empty land that could work for future parks, flat and near roads, at least 10 hectares big.
- **NON-IT (Green):** Homes, farms, forests, or protected spots where IT parks shouldn't go, like Cubbon Park.

Why Bangalore is Chosen ?

Bangalore is India's Silicon Valley, home to major IT hubs like Electronic City, Whitefield, Manyata Tech Park, and the Outer Ring Road corridor. It has a booming tech industry with companies like Infosys, Wipro, TCS, Google, and Microsoft. The city faces rapid urbanization and IT expansion, making it ideal for this project. Plus, there's lots of free satellite data available for analysis, and it's a real-world problem here with overcrowding and traffic issues.

Data Collection Resources

- **Raster Data:** USGS, WorldPop.
- **Vector Data:** OpenStreetMap.

QGIS Plugins Used

- **Land Cover Downloader:** For downloading land cover data.
- **Quick Map Services:** To add basemaps and satellite imagery layers.
- **OSM Standard:** For standard OpenStreetMap data integration.

Raster Data (Bangalore)

Satellite Imagery Data

- **B2.TIF (Band 2 - Blue)**

Captures blue wavelength for water and atmospheric analysis.

Used to filter water bodies and improve spectral index accuracy.

Source: USGS Sentinel-2

- **B3.TIF (Band 3 - Green)**

Detects vegetation and green areas in the city.

Essential for creating RGB composites and complementing NDVI calculations.

Source: USGS Sentinel-2

- **B4.TIF (Band 4 - Red)**

Primary band for NDVI and NDBI calculations to identify vegetation and built-up areas.

Distinguishes IT parks from open land and green spaces.

Source: USGS Sentinel-2

- **Bangalore_RGB.tif**

True-color composite for visual validation of IT zones and reference mapping.

Helps analysts verify automated classifications against actual satellite imagery.

Source: Derived from B2, B3, B4 combined

Raster Data (Bangalore)

Derived Analysis Data

DEM.tif

- Provides elevation data for slope and aspect derivation.
- Essential for identifying flat terrain suitable for IT park development.
- **Source:** USGS SRTM

Slope.tif

- Identifies flat areas where IT parks can be built without terracing.
- Scoring criteria: flatter = higher suitability score.
- **Source:** Derived from DEM

Aspect.tif

- Analyzes slope direction for infrastructure planning and environmental assessment.
- Helps understand terrain orientation relative to development.
- **Source:** Derived from DEM

HillShade.tif

- Creates shaded relief visualization for better map readability.
- Overlay on zone maps for clearer terrain understanding.
- **Source:** Derived from DEM

Land_Cover_Raster.tif

- Classifies land use into vegetation, built-up, water, and agriculture categories.
- Directly identifies potential IT zones (built-up/open) and non-IT zones (green/water).
- **Source:** Land Cover over AOI Plugin

population_bangalore.tif

- Measures urbanization intensity and population concentration.
- Indicates existing IT corridors and development centers for proximity analysis.
- **Source:** USGS WorldPop

Vector Data (Bangalore)

Infrastructure & Transportation Data

- **banglore_bus_stop**
 - Marks public transport accessibility points for IT zones.
 - Connectivity scoring: proximity to bus stops increases suitability score.
 - **Source:** OpenStreetMap
- **banglore_railway**
 - Shows railway network for transportation connectivity assessment.
 - Major connectivity factor for IT park suitability and employee accessibility.
 - **Source:** OpenStreetMap
- **banglore_railway_stations**
 - Identifies major transit hubs near potential IT sites.
 - Zones within 5km of stations get priority suitability scoring.
 - **Source:** OpenStreetMap
- **banglore_parking (Large & Small)**
 - Detects parking areas of varying sizes indicating existing IT parks and commercial zones.
 - Large parking areas indicate IT zones; smaller facilities help distinguish commercial from IT areas.
 - **Source:** OpenStreetMap
- **banglore_power_line & power_pole**
 - Confirms utility infrastructure availability for IT parks.
 - Critical infrastructure requirement: scored in suitability assessment.
 - **Source:** OpenStreetMap
- **banglore_outline**
 - Defines study area boundary for raster clipping and analysis.
 - Ensures analysis focuses only on Bangalore city limits.
 - **Source:** OpenStreetMap
- **Hebbal_roads, Indiranagar_roads, Kalkere_roads, Marathahalli_roads**
 - Maps road networks in specific IT corridors for connectivity analysis.
 - Distance-to-road scoring: areas within 1km of roads get higher suitability.
 - **Source:** OpenStreetMap

Vector Data (Bangalore)

Land Use & Environmental Data

banglore_residential

- Identifies residential zones as non-IT areas (exclusion criteria).
- Applied as negative weighting in potential zone identification.
- **Source:** OpenStreetMap

banglore_commercial

- Detects commercial clusters indicating urban development centers.
- May include IT parks; used for zone refinement and validation.
- **Source:** OpenStreetMap

banglore_industrial

- Maps industrial zones, some containing IT parks (SEZ or IT parks).
- Used to identify emerging IT zones adjacent to established ones.
- **Source:** OpenStreetMap

banglore_farmland

- Locates agricultural open land suitable for IT park development.
- Primary source for Potential IT (Orange) zone identification.
- **Source:** OpenStreetMap

banglore_parks (Large & Small)

- Identifies parks and green patches as public amenities and recreation areas.
- Non-IT zone markers; proximity increases quality-of-life scoring for potential zones.
- **Source:** OpenStreetMap

banglore_green_space

- Marks protected green areas as non-IT zones (Green classification).
- Excluded from development suitability; scored as non-buildable areas.
- **Source:** OpenStreetMap

banglore_water (Natural & Artificial)

- Identifies water bodies and water sources for analysis.
- Non-IT zone marker; locates flood-prone areas and water access for sustainability scoring.
- **Source:** OpenStreetMap

Hebbal_buildings, Indiranagar_buildings, Kalkere_buildings, Marathahalli_buildings

- Analyzes building footprints and urban density in IT corridors.
- High building density in specific patterns indicates established/emerging IT zones.
- **Source:** OpenStreetMap

Future Data to Collect

Raster Data (Derived Indices)

- **NDVI** - Vegetation detection. Derive: $(B8-B4)/(B8+B4)$ using QGIS Raster Calculator. Source: USGS Sentinel-2
- **NDBI** - Built-up areas. Derive: $(B11-B8)/(B11+B8)$ using QGIS Raster Calculator. Source: USGS Sentinel-2
- **NDWI** - Water bodies. Derive: $(B3-B8)/(B3+B8)$ using QGIS Raster Calculator. Source: USGS Sentinel-2
- **Urban Heat Island (LST)** - Thermal mapping. Download from Google Earth Engine or process Landsat 8 Band 10/11. Source: USGS Landsat 8

Vector Data (Proximity/Distance Layers)

- **Road Proximity Buffers** - 1km, 2km, 5km zones around roads. Use QGIS Distance to Hub tool. Source: OpenStreetMap
- **Metro/Rail Proximity Buffers** - 500m, 1km, 2km, 5km zones. Use QGIS Distance tool. Source: OpenStreetMap
- **Airport Proximity** - Distance to airport. Buffer 15-20km zone. Source: OpenStreetMap
- **IT Park Proximity** - Distance to existing IT parks. Digitize known parks + buffer 2-5km. Source: KIADB/Manual

THANK YOU ...